

2 Overview of Bluetooth Wireless Technology

Bluetooth is an open standard for short-range radio frequency (RF) communication. Bluetooth is used primarily to establish wireless personal area networks (WPANs). Bluetooth has been integrated into many types of business and consumer devices, including cell phones, laptops, automobiles, printers, keyboards, mice, headsets, and, more recently, medical devices and personal devices (such as smart watches, music speakers, home appliances, fitness monitors, and trackers). This allows users to form ad hoc networks between a wide variety of devices to transfer voice and data. Bluetooth is a low-cost, low-power technology that provides a mechanism for creating small wireless networks on an ad hoc basis, known as *piconets*.² A piconet is composed of two or more Bluetooth devices in close physical proximity that operate on the same channel using the same frequency hopping sequence. An example of a piconet is a connection between a cell phone and a headset using Bluetooth wireless technology.

Bluetooth piconets are often established on a temporary and changing basis, which offers communications flexibility and scalability between mobile devices. Some key benefits of Bluetooth are—

- **Cable replacement.** Bluetooth replaces a variety of cables, such as those traditionally used for peripheral devices (e.g., mouse and keyboard connections), printers, and wired headsets and earbuds that interface with desktops, laptops, cell phones, etc.
- **Ease of file sharing.** A Bluetooth-enabled device can form a piconet to support file sharing capabilities with other Bluetooth devices, such as laptops.
- **Wireless synchronization.** Bluetooth can provide automatic synchronization between Bluetooth-enabled devices. For example, Bluetooth allows synchronization of contact information between smartphones and automobiles.
- **Internet connectivity.** A Bluetooth device with Internet connectivity can share that access with other Bluetooth devices. For example, a laptop can use a Bluetooth connection to leverage the personal hotspot capability of a smartphone to provide Internet access to the laptop.

Bluetooth was originally conceived by Ericsson in 1994. Ericsson, IBM, Intel, Nokia, and Toshiba formed the Bluetooth Special Interest Group (SIG), a not-for-profit trade association developed to drive development of Bluetooth products and serve as the governing body for Bluetooth specifications.³ Bluetooth is standardized within the IEEE 802.15 Working Group for Wireless Personal Area Networks that formed in 1999 as IEEE 802.15.1-2002.⁴

This section provides an overview of Bluetooth, including frequency and data rates, range, and architecture.

² As discussed in Section 2.2, the term “piconet” applies to both ad hoc and infrastructure Bluetooth networks.

³ The Bluetooth SIG website (<https://www.bluetooth.com/>) is a resource for Bluetooth related information and provides numerous links to other sources of information.

⁴ For more information, see the IEEE website at <http://grouper.ieee.org/groups/802/15/>.

2.1 Bluetooth Wireless Technology Characteristics

Bluetooth operates in the unlicensed 2.4000 gigahertz (GHz) to 2.4835 GHz Industrial, Scientific, and Medical (ISM) frequency band. Numerous technologies operate in this band, including the IEEE 802.11b/g/n wireless local area network (WLAN) standard, making it somewhat crowded from the standpoint of the volume of wireless transmissions. Bluetooth employs frequency hopping spread spectrum (FHSS) technology for transmissions. FHSS reduces interference and transmission errors but provides minimal transmission security.

With FHSS technology, communications between Bluetooth Basic Rate (BR)/EDR devices use 79 different 1 megahertz (MHz) radio channels by hopping (i.e., changing) frequencies about 1600 times per second for data/voice links and 3200 times per second during page and inquiry scanning. A channel is used for a very short period (e.g., 625 μ s for data/voice links), followed by a hop to another channel designated by a pre-determined pseudo-random sequence; this process is repeated continuously in the frequency hopping sequence.

Bluetooth low energy communication uses the same frequency range as BR/EDR devices but splits it instead into 40 channels of 2 MHz width. Three of these channels are used for advertising (broadcasting data and for connection setup) and the other 37 are data channels. These 40 channels, combined with a time division multiple access (TDMA) scheme, provide the two multiple access schemes for the low energy feature of Bluetooth. A polling scheme is used in which the first device sends a packet at a predetermined time and a corresponding device responds after a predetermined interval. These exchanges of data are known as either Advertising or Connection Events.

Bluetooth also provides for radio link power control, which allows devices to negotiate and adjust their radio power according to signal strength measurements. Each device in a Bluetooth network can determine its received signal strength indication (RSSI) and request that the other network device adjust its relative radio power level (i.e., incrementally increase or decrease the transmission power). This is performed to conserve power and/or to keep the received signal characteristics within a preferred range.

The combination of a frequency hopping scheme and radio link power control provides Bluetooth with some additional, albeit limited, protection from eavesdropping and malicious access. The frequency-hopping scheme, primarily a technique to avoid interference, makes it slightly more difficult for an adversary to locate and capture Bluetooth transmissions than to capture transmissions from fixed-frequency technologies, like those used in IEEE 802.11b/g. Research has shown that the Bluetooth frequency hopping sequence for an active piconet can be determined using relatively inexpensive hardware and free open source software.⁵

The range of Bluetooth BR/EDR devices is characterized by three classes that define power management. Table 2-1 summarizes the classes, including their power levels in milliwatts (mW) and decibels referenced to one milliwatt (dBm), and their operating ranges in meters (m).⁶ Most small, battery-powered devices are Class 2, while Class 1 devices are typically universal serial bus (USB)

⁵ Spill, Dominic and Bittau, Andrea, *BlueSniff: Eve meets Alice and Bluetooth*, 2007.

⁶ The ranges listed in Table 2-1 are the designed operating ranges. Attackers may be able to intercept communications at significantly larger distances, especially if they use high-gain antennas and high-sensitivity receivers.

adapters for desktops and laptops, as well as access points and other mains powered devices. Many Bluetooth low energy devices are designed to run on very small batteries for a long period of time.

Table 2-1. Bluetooth Device Classes of Power Management

Type	Power	Max Power Level	Designed Operating Range	Sample Devices
Class 1	High	100 mW (20 dBm)	Up to 100 m (328 feet)	USB adapters, access points
Class 1.5 (low energy) ⁷	Med-High	10 mW (10 dBm)	Up to 30 m (100 feet), but typically 5 m (16 feet)	Beacons, wearable sensors
Class 2	Medium	2.5 mW (4 dBm)	Up to 10 m (33 feet)	Mobile devices, Bluetooth adapters, smart card readers
Class 3	Low	1 mW (0 dBm)	Up to 1 m (3 feet)	Bluetooth adapters

To allow Bluetooth devices to find and establish communication with each other, discoverable and connectable modes are specified. A device in *discoverable mode* periodically monitors an inquiry scan physical channel (based on a specific set of frequencies) and responds to an inquiry on that channel with its device address, local clock (counter) value, and other characteristics needed to page and subsequently connect to it. A device in *connectable mode* periodically monitors its page scan physical channel and responds to a page on that channel to initiate a network connection. The frequencies associated with the page scan physical channel for a device are based on its Bluetooth address. Therefore, knowing a device's address and local clock⁸ is important for paging and subsequently connecting to the device.

The following sections cover Bluetooth BR/EDR/HS data rates, low energy technology, and dual mode devices.

2.1.1 Basic, Enhanced, and High Speed Data Rates

Bluetooth devices can support multiple data rates using native Bluetooth and alternate Media Access Control (MAC) and Physical (PHY) Layers. Bluetooth specifications are designed to be backward compatible; a later specification device that supports higher data rates also supports the lower data rates supported by earlier specification devices (e.g., an EDR device also supports rates specified for BR devices). The following sections provide an overview for Bluetooth and alternate MAC/PHYs, as well as associated data rates and modulation schemes.

2.1.1.1 Basic Rate/Enhanced Data Rate

Bluetooth versions 1.1 and 1.2 only support transmission speeds of up to 1 megabit per second (Mbps), which is known as Basic Rate (BR), and can achieve payload throughput of approximately

⁷ Bluetooth Core Specification Addendum (CSA) v5 introduced Power Class 1.5 (10mW), which was the maximum output power of Bluetooth low energy 4.0-4.2 devices. CSA v5 also increased the maximum output power for low energy devices to 100mW as long as local regulatory bodies allow it.

⁸ Having a remote device's clock information is not needed to make a connection, but it will speed up the connection process.

720 kilobits per second (kbps). Introduced in Bluetooth version 2.0, EDR specifies data rates up to 3 Mbps and throughput of approximately 2.1 Mbps.

BR uses Gaussian Frequency-Shift Keying (GFSK) modulation to achieve a 1 Mbps data rate. EDR uses $\pi/4$ rotated Differential Quaternary Phase Shift Keying (DQPSK) modulation to achieve a 2 Mbps data rate, and 8 Phase Differential Phase Shift Keying (8DPSK) to achieve a 3 Mbps data rate.

Note that EDR support is not required for devices compliant with the Bluetooth 2.0 specification or later. Therefore, there are devices on the market that are “Bluetooth 2.0 compliant” versus “Bluetooth 2.0 + EDR compliant.” The former are devices that support required version 2.0 features but only provide the BR data rate.

2.1.1.2 High Speed with Alternate MAC/PHY

Introduced in the Bluetooth 3.0 + HS specification, devices can support faster data rates by using Alternate MAC/PHYs (AMP). This is known as Bluetooth high speed technology.

In the Bluetooth 3.0 + HS specification, IEEE 802.11-2007 was introduced as the first supported AMP. IEEE 802.11-2007 is a rollup of the amendments IEEE 802.11a through 802.11j. For the 802.11 AMP, IEEE 802.11g PHY support is mandatory, while IEEE 802.11a PHY support is optional. The 802.11 AMP is designed to provide data rates up to 24 Mbps using Orthogonal Frequency-Division Multiplexing (OFDM) modulation.

Note that this AMP is IEEE 802.11 compliant but not Wi-Fi compliant. Therefore, Wi-Fi Alliance specification compliance is not required for Bluetooth 3.0 + HS devices.

2.1.2 Low Energy

Bluetooth low energy was introduced in the Bluetooth 4.0 specification and updated in 4.1 and 4.2. Formerly known as “Wibree” and “Ultra Low Power Bluetooth,” low energy is primarily designed to bring Bluetooth to coin cell battery-powered devices such as medical devices and other sensors. The key technology goals of Bluetooth low energy (compared with Bluetooth BR/EDR) include lower power consumption, reduced memory requirements, efficient discovery and connection procedures, short packet lengths, and simple protocols and services.

Table 2-2 provides the key technical differences between BR/EDR and low energy.

Table 2-2. Key Differences Between Bluetooth BR/EDR and Low Energy

Characteristic	Bluetooth BR/EDR		Bluetooth Low Energy	
	Prior to 4.1	4.1 onwards	Prior to 4.2	4.2 onwards
RF Physical Channels	79 channels with 1 MHz channel spacing		40 channels with 2 MHz channel spacing	
Discovery/Connect	Inquiry/Paging		Advertising	
Number of Piconet Slaves	7 (active)/255 (total)		Unlimited	
Device Address Privacy	None		Private device addressing available	
Max Data Rate	1–3 Mbps		1 Mbps via GFSK modulation	
Pairing Algorithm	Prior to 2.1: E21/E22/SAFER+	P-256 Elliptic Curve, HMAC-SHA-256	AES-128	P-256 Elliptic Curve, AES-CMAC
	2.1-4.0: P-192 Elliptic Curve ⁹ , HMAC-SHA-256			
Device Authentication Algorithm	E1/SAFER	HMAC-SHA-256	AES-CCM ¹⁰	
Encryption Algorithm	E0/SAFER+	AES-CCM	AES-CCM	
Typical Range	30 m		50 m	
Max Output Power	100 mW (20 dBm)		10 mW (10 dBm) ¹¹	

2.1.3 Dual Mode Devices (Concurrent Low Energy & BR/EDR/HS Support)

A Bluetooth 4.0 or later device may support both BR/EDR/HS and low energy as a “dual mode” Bluetooth device. An example is a cell phone that uses an EDR link to a Bluetooth headset and a concurrent low energy link to a sensor that unlocks and starts the user’s automobile. Figure 2-1 shows the device architecture for Bluetooth 4.x devices, and includes BR/EDR, HS, and low energy technologies. New terms included in the figure related to security are discussed in subsequent sections.

⁹ For more information about P-192, see FIPS 186-4, *Digital Signature Standards (DSS)*.

¹⁰ There is no dedicated device authentication algorithm in low energy. Encrypting the link also successfully authenticates the remote device.

¹¹ Core Specification Addendum 5 (CSA5) changed this to 100 mW (20 dBm) as long as the regulatory bodies permit it.

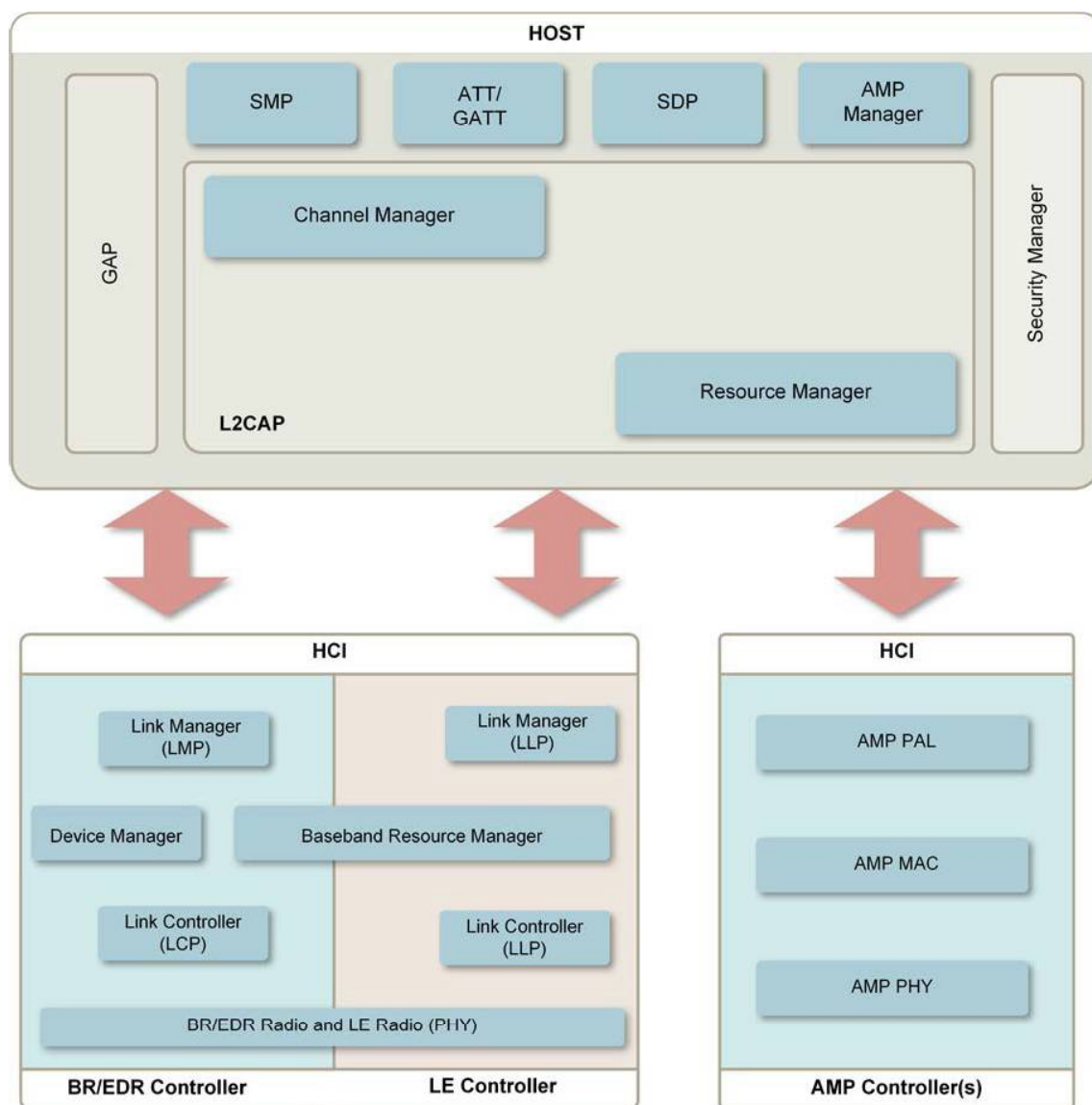


Figure 2-1. Bluetooth 4.x Device Architecture

2.2 Bluetooth Architecture

Bluetooth permits devices to establish ad hoc networks. Ad hoc networks allow easy connection establishment between devices in the same physical area (e.g., the same room) without the use of any infrastructure devices. A Bluetooth client is simply a device with a Bluetooth radio and software incorporating the Bluetooth protocol stack and interfaces.

The Bluetooth specification provides separation of duties for performing stack functions between a host and a controller. The host is responsible for the higher layer protocols, such as Logical Link Control and Adaptation Protocol (L2CAP) and Service Discovery Protocol (SDP). The host functions are performed by a computing device like a laptop or smartphone. The controller is responsible for the lower layers, including the Radio, Baseband, and Link Control/Management.

The controller functions are performed by an integrated or external (e.g., USB) Bluetooth adapter. The host and controller send information to each other using standardized communications over the Host Controller Interface (HCI). This standardized HCI allows hosts and controllers from different product vendors to interoperate. In some cases, the host and controller functions are integrated into a single device; Bluetooth headsets are a prime example.

Figure 2-2 depicts the basic Bluetooth network topology. In a piconet one device serves as the master, with all other devices in the piconet acting as slaves. BR/EDR piconets can scale to include up to 7 active slave devices and up to 255 inactive slave devices. Bluetooth low energy (see Section 2.1.2) allows an unlimited number of slaves, which is known as the low energy Peripheral role, with the master being the low energy Central role. The other two low energy device roles, Broadcaster and Observer, are discussed below in this section.

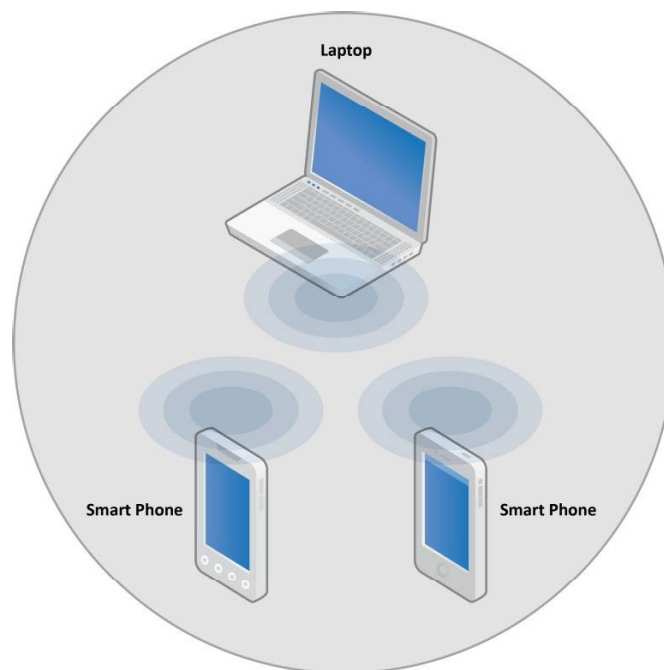


Figure 2-2. Bluetooth Ad Hoc Topology

The master device controls and establishes the network, including defining the network's frequency hopping scheme. Although only one device can serve as the master for each piconet, time division multiplexing (TDM) allows a slave in one piconet to act as the master for another piconet simultaneously, thus creating a chain of networks.¹² This chain, called a *scatternet*, allows networking of several devices over an extended distance in a dynamic topology that can change during any given session. As a device moves toward or away from the master device the topology may change, along with the relationships of the devices in the immediate network. Figure 2-3 depicts a scatternet that involves three piconets.

¹² Note that a particular device can only be the master of one piconet at any given time.

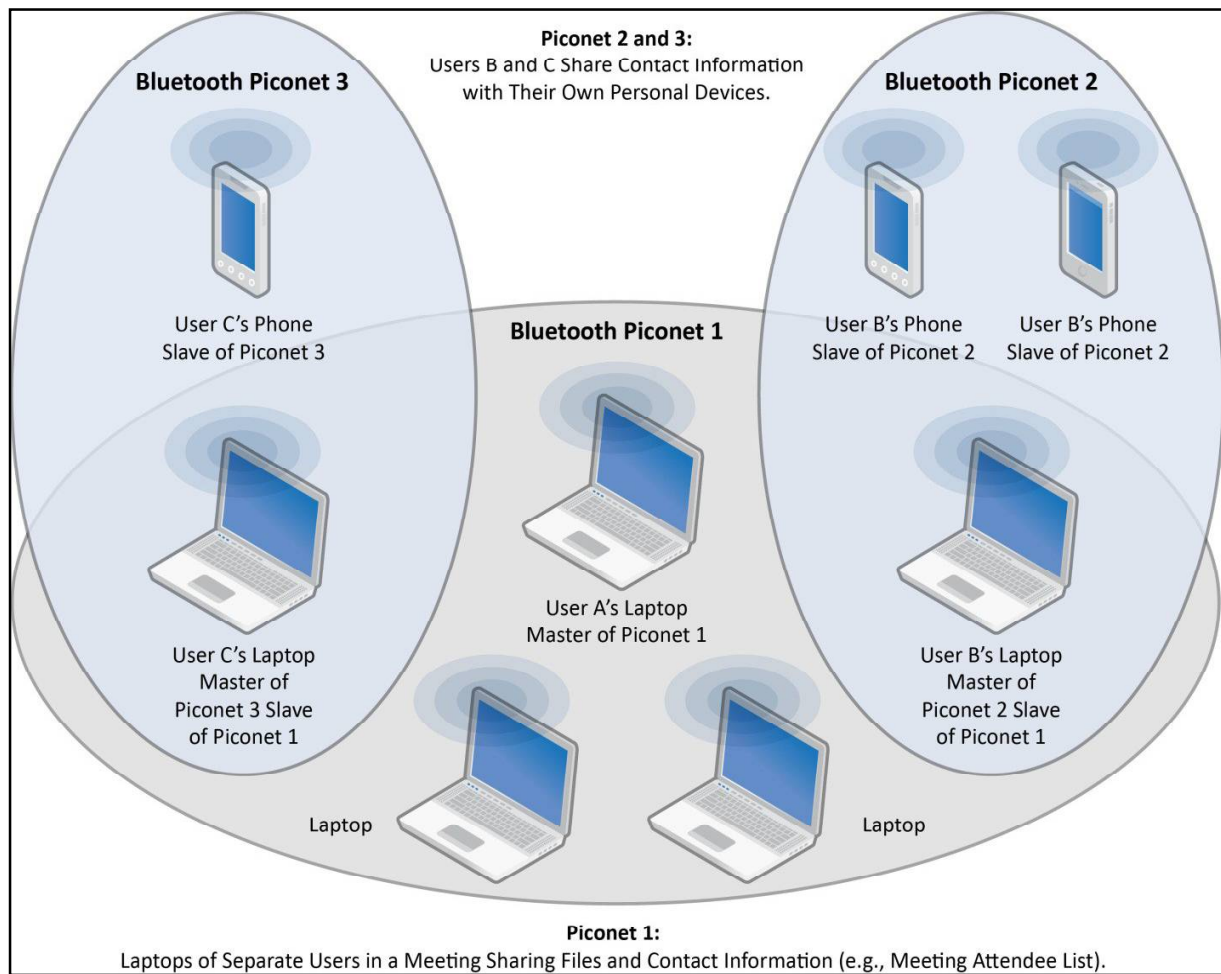


Figure 2-3. Bluetooth Networks (Multiple Scatternets)

The Bluetooth core protocols provide no multi-hop network routing capabilities for devices involved in scatternets. For example, in Figure 2-3, User C's phone in Piconet 3 cannot communicate with User B's phones in Piconet 2 without establishing an additional piconet between them.

Scatternets are supported by both BR/EDR and low energy technologies.

Low energy functionality also supports a connectionless broadcast architecture where Broadcasters (low energy device role) periodically send data, and Observers (low energy device role) listen and consume that data. This allows a device to transmit data to more than one peer at a time. The broadcasting function is a subset of the Advertising capability used in the low energy connection architecture.